MODEL IDENTIFICATION AND DATA ANALYSIS – Part 1 (Automation&Control Engineering), A.Y. 2021/2022

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- Write the solutions in the blank areas (use the back of the page, if needed)
- The number of pages is 4. Additional papers will not be considered.
- Clarity, order and precision are strongly considered for the final evaluation.

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1. [Linear regression]

Recall that the objective function for L2-regularized linear regression is

$$\mathrm{J}(\mathrm{w}) = \|\mathrm{Xw} - \mathrm{y}\|_2^2 + \lambda\|\mathrm{w}\|_2^2$$

where X is the design matrix (the rows of X are the data points). The global minimizer of J is given by:

$$\mathrm{w}^* = \left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)^{-1}\mathrm{X}^\mathrm{T}\mathrm{Y}$$

As an alternative method to minimize J, consider running Newton's method (i.e., the gradient descent formula with the step size substituted by the inverse of the hessian of J). Let w_0 be an arbitrary initial guess for Newton's method. Show that w_1, the value of the weights after one iteration of Newton's method, is equal to w^*.

a. Explain why L_2-regularized regression may lead to more parsimonious models than simple linear regression.
b. Describe one way to tune λ.

Solutions

a. As indicated also in the text, the Newton's Method for Optimization is

$$\mathrm{w}_1 = \mathrm{w}_0 - \left(\left.\frac{\partial^2 J}{\partial w^2}\right|_{\mathrm{w}_0}\right)^{-1} \left.\frac{\partial J}{\partial w}\right|_{\mathrm{w}_0}$$

The gradient and the hessian of $\mathrm{J}(\mathrm{w}) = \|\mathrm{Xw} - \mathrm{y}\|_2^2 + \lambda\|\mathrm{w}\|_2^2$ are as follows:

$$\frac{\partial J}{\partial w} = 2\mathrm{X}^\mathrm{T}\mathrm{X}w - 2\mathrm{X}^\mathrm{T}\mathrm{Y}w + 2\lambda w = 2\left[\left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)w - \mathrm{X}^\mathrm{T}\mathrm{Y}\right]$$

$$\frac{\partial^2 J}{\partial w^2} = 2\left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)$$

We initialize w_0 to some value. Note that this won't matter. Plugging this in, we have

$$\mathrm{w}_1 = \mathrm{w}_0 - \left(\left.\frac{\partial^2 J}{\partial w^2}\right|_{\mathrm{w}_0}\right)^{-1} \left.\frac{\partial J}{\partial w}\right|_{\mathrm{w}_0} = \mathrm{w}_0 - \left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)^{-1}\left[\left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)\mathrm{w}_0 - \mathrm{X}^\mathrm{T}\mathrm{Y}\right] = \mathrm{w}_0 - \mathrm{w}_0 + \left(\mathrm{X}^\mathrm{T}\mathrm{X} + \lambda\mathrm{I}\right)^{-1}\mathrm{X}^\mathrm{T}\mathrm{Y}$$

Thus, $\mathrm{w}_1 = \mathrm{w}^*$.

b. See the lecture notes (chapter 09).
c. See the lecture notes (chapter 09).

2. [Logistic regression]

Gareth is from New Zealand where the national sports passion is rugby. The national rugby team is called the All Blacks (they wear black) and their main rivals are Australia (the Wallabies) and South Africa (the Springboks). Gareth realizes that what he really cares about is whether the All Blacks win or not. Therefore, he decides to perform a logistic regression with the response variable, Y, being whether or not the All Blacks win (Y = 1 if they win and 0 if they lose). The regressors are:

- AB Win%: the percent of the previous 10 games that the All Blacks had won going into the game in question,
- Opp Win%: same definition for the opponents last 10 games,
- Home: an indicator variable with 1 corresponding to an All Blacks home game and 0 an away game,
- Temperature: the temperature at which the game was played,

a. Describe the logistic regression model and the maximum likelihood approach for its calibration.

b. The obtained coefficients for each regressor are as follows.

	BIAS	AB Win%	Opp Win%	Home	Temperature
Coeff	*-25.30*	*0.466*	*-0.170*	*1.45*	*0.115*

What does the coefficient for Temperature tell us about the relationship between temperature and the probability that the All Blacks win?

c. Estimate the probability of the New Zealand All Blacks winning a game against South Africa played in South Africa at 50 degree temperatures where both teams have a winning percentage of 70.

d. The coefficient for the Home variable seems to indicate that the All Blacks are more likely to win at home than on the road. However, somewhat surprisingly, the All Blacks turn out to win more games on the road than at home! Assuming that the model is correct (i.e., there are no important variables missing from the model or violations of the basic assumptions etc.) and the coefficient estimates are exactly correct, how could the coefficient for Home be positive even though the All Blacks win more games on the road?

Solutions

a. See the lecture notes (chapter 04).

b. The coefficient for temperature is positive. Hence, holding all other variables fixed, on warmer days the All Blacks are more likely to win than on colder days.

c. The logistic formula is

$$p = \frac{e^s}{1+e^s} = \frac{e^{w_0+w_1X_1+\ldots+w_5X_5}}{1+e^{w_0+w_1X_1+\ldots+w_5X_5}} = \frac{e^{-25.3+0.466(70)-0.170(70)+1.45(0)+0.115(50)-0.245(0)}}{1+e^{w_0+w_1X_1+\ldots+w_5X_5}} = 0.763$$

The All Black's chances of winning the game are quite good (76.3%)!

d. There are at least a couple of possible explanations for this effect. The key here is that the Home coefficient being positive only tells us that if all the other variables are the same then the All Blacks are more likely to win at home than on the road. However, the other variables may not all be the same. For example, suppose that the All Blacks always play better teams (as measured by Opp Win%) at home and worse teams on the road. Then this might negate the otherwise positive effect of being at home. Another possibility is temperature. Suppose that the All Blacks home games tend to be colder than their away games. Then, since they prefer playing in warmer temperatures, they could well end up winning more games on the road.

3. [Principal component analysis]

a. Describe the SVD algorithm used for principal component analysis and explain when this might be useful for dimensionality reduction.

b. If you perform an arbitrary rigid rotation of the sample points as a group in feature space before performing PCA, does the largest eigenvalue of the sample covariance matrix change? Explain your answer.

Solutions

a. See the lecture notes (chapter 12).

b. No. In fact, rotating the sample points rotates the principal components, but it does not change the variance along each of those (rotated) component directions.

4. **[Learning SPs]**
   Describe the sample estimators for mean, covariance function and spectral density of a stationary stochastic process. What are the problems related to the estimation of the spectrum? How can they be solved?
   Can such estimators be used also when the process is characterized by a seasonal behavior?

Solutions

See the lecture notes (chapters 15 and 16).